

Adaboost

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Adaboost

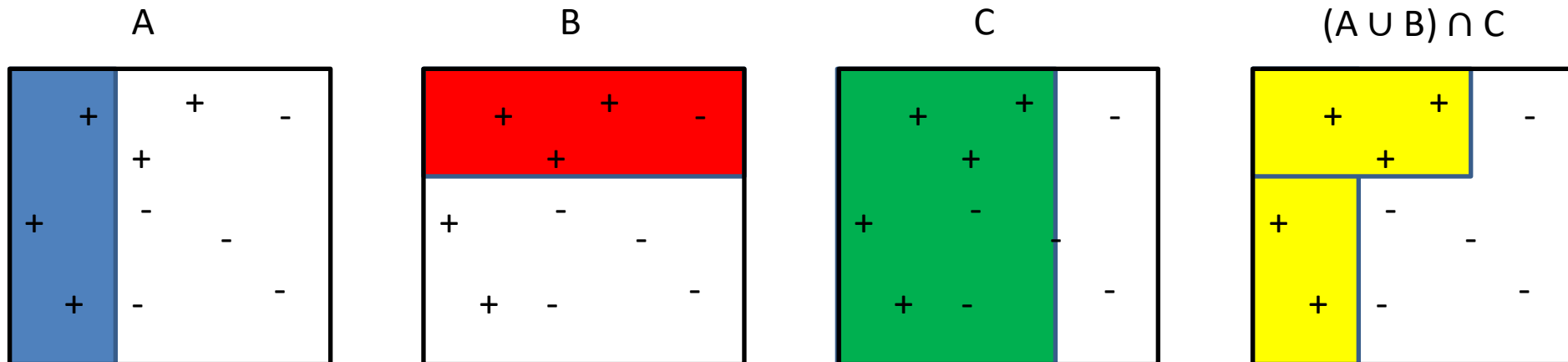
- Suppose we have many **weak learners**
 - Their error is better than .5
 - but maybe not much better
- Example: Spam detection
 - Contains “buy now” in title
 - Contains “medz” in title
 - Contains link in body
 - Contains prices in body

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- Suppose we have many **weak learners**
 - Their error is better than .5
 - but maybe not much better
- Example: betting on horses
 - Win streak
 - Winning percentage
 - Pedigree

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- Can we join together weak rules to derive a stronger rule?
- Approach:
 - Give each rule a weight of importance
 - Take a weighted vote



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- How many weak learners should we choose?
- Bias-variance tradeoff
 - Bias: Empirical error on sample
 - Variance: Complexity of solution (VC-dimension)
- Overfitting:
 - Low empirical error but high VC-dimension
 - Bad on world (true error)

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- Input:
 - Set S of points, $x_i \in S$, with labels y_i
 - Number of iterations k .
 - A set $\mathbf{H}$ of T (weak) classifiers, $h_j: S \rightarrow \{-1, 1\}^{|S|}$
- Output: a weight α_j for each classifier h_j .
- Final decision function:
 - $F(x) = \sum_{t=1}^T \alpha_t h_t(x)$
 - $H(x) = \text{sign}[F(x)]$

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- Algorithm:

- 1. Initialize point weights $D_1(x_i) = \frac{1}{n}$
- 2. For iteration $t=1,\dots,k$
- 3. Compute weighted error for each $h \in H$:
$$\epsilon_t(h) = \sum_{i=1}^n D_t(x_i)[h(x_i) \neq y_i]$$
- 4. Select classifier with min weighted error
$$h_t = \operatorname{argmin}_h \epsilon_t(h)$$
- 5. Set classifier weight α_t based on its error
$$\alpha_t = \frac{1}{2} \ln \frac{1-\epsilon_t(h_t)}{\epsilon_t(h_t)}$$
- 6. Update point weights

$$D_{t+1}(x_i) = \frac{1}{Z_t} D_t(x_i) e^{-\alpha_t h_t(x_i) y_i}$$

where Z_t is a normalizing constant giving $\sum_i D_{t+1}(x_i) = 1$

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If $\epsilon_t(h_t)$ is close to 0,
weight of classifier is high
If $\epsilon_t(h_t)$ is close to $\frac{1}{2}$,
weight of classifier is low

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If $h_t(x_i) = y_i$, then
 $h_t(x_i)y_i = 1$, and point
weight decreases
If $h_t(x_i) \neq y_i$, then
 $h_t(x_i)y_i = -1$, and point
weight increases

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- Analysis:

- $$\begin{aligned} D_{t+1}(x_i) &= \frac{1}{Z_t} D_t(x_i) e^{-\alpha_t h_t(x_i) y_i} \\ &= \frac{1}{Z_t Z_{t-1}} D_{t-1}(x_i) e^{-y_i (\alpha_t h_t(x_i) + \alpha_{t-1} h_{t-1}(x_i))} \\ &= \frac{1}{Z_t Z_{t-1} \cdots Z_1} D_1(x_i) e^{-y_i \sum_{j=1}^t \alpha_j h_j(x_i)} \\ &= \frac{1}{Z} \cdot \frac{1}{n} e^{-y_i F(x_i)} \end{aligned}$$

- Here Z is the normalization constant

$$Z = Z_t Z_{t-1} \cdots Z_1 = \frac{1}{n} \sum_{i=1}^n e^{-y_i F(x_i)}$$

Where the second equality follows from

$$1 = \sum_{i=1}^n D_{t+1}(x_i)$$

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- Claim:
 - $\text{Empirical_Error}(H) \leq Z$ Good! Decreases each round
 - $\frac{1}{n} \sum_{i=1}^n [H(x_i) \neq y_i] \leq \frac{1}{n} \sum_{i=1}^n e^{-y_i F(x_i)}$
- Proof:
 - $F(x_i) = \text{sign}(F(x_i)) |F(x_i)| = H(x_i) |F(x_i)|$
 - If $H(x_i) \neq y_i$
 - LHS: $\text{Error}(H(x_i)) = 1$
 - RHS: $e^{-y_i F(x_i)} = e^{-y_i H(x_i) |F(x_i)|} = e^{-|F(x_i)|} \geq 1$
 - If $H(x_i) = y_i$
 - LHS: $\text{Error}(H(x_i)) = 0$
 - RHS: $e^{-y_i F(x_i)} = e^{-y_i H(x_i) |F(x_i)|} = e^{-|F(x_i)|} \geq 0$

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- So we want to minimize Z
 - How do we do this?
- Note: $Z_t = \sum_{i=1}^n D_t(x_i) e^{-y_i \alpha_i h(x_i)}$
$$= \sum_{x_i \in A} D_t(x_i) e^{-\alpha_t} + \sum_{x_i \in \bar{A}} D_t(x_i) e^{\alpha_t}$$
where A are the correctly classified points

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- Take derivative of Z_t :

- $0 = Z'_t = -\sum_{x_i \in A} D_t(x_i) e^{-\alpha_t} + \sum_{x_i \in \bar{A}} D_t(x_i) e^{\alpha_t}$

- So we want $\sum_{x_i \in A} D_t(x_i) = \sum_{x_i \in \bar{A}} D_t(x_i) e^{2\alpha_t}$

$$1 - \epsilon_t(h_t)$$

$$\epsilon_t(h_t)$$

- Solve for α_t : $\alpha_t = \frac{1}{2} \ln \frac{1 - \epsilon_t(h_t)}{\epsilon_t(h_t)}$
 - Exactly the value Adaboost takes for α_t
 - So Adaboost maximizes the decrease in Z at every step

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- Plug value of α_t into Z_t :
- $$Z_t = \sum_{x_i \in A} D_t(x_i) e^{-\alpha_t} + \sum_{x_i \in \bar{A}} D_t(x_i) e^{\alpha_t}$$
$$= (1 - \epsilon_t(h_t)) \sqrt{\frac{\epsilon_t(h_t)}{1 - \epsilon_t(h_t)}} + \epsilon_t(h_t) \sqrt{\frac{1 - \epsilon_t(h_t)}{\epsilon_t(h_t)}}$$
$$= 2\sqrt{\epsilon_t(h_t)(1 - \epsilon_t(h_t))}$$
- Substitute $\beta_t = \frac{1}{2} - \epsilon_t(h_t)$ $\beta_t \in [0, \frac{1}{2}]$
 - $z_t = \sqrt{1 - 4\beta_t^2} \leq e^{-2\beta_t^2}$
- So: $\text{Error}(H) \leq Z \leq e^{-2 \sum_{t=1}^T \beta_t^2}$
 - Decrease faster with smaller error